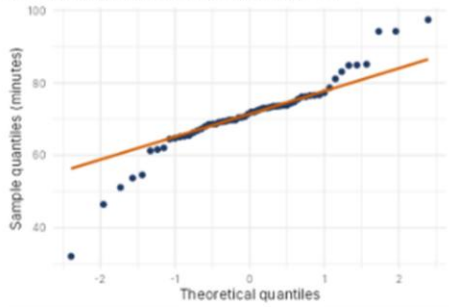
DSA06 - Data Handling and Visualization | Activity 5: Normality Detection using Q-Q Plots | Slot D

R.No: 16 Reg. No: 192424097 Name: K DEEKSHA SREE

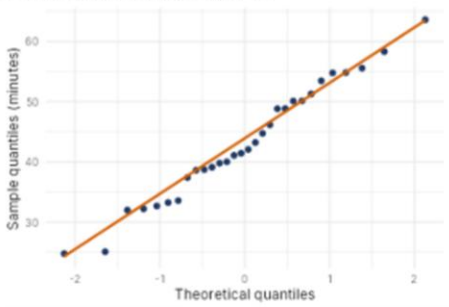
Each chart is a normal Q-Q plot made in R (stat_qq and stat_qq_line; red line = perfect normal fit). Part A (16 marks): for every chart decide whether the data are normal and name the shape from this list: Normal, Right-skewed, Left-skewed, Heavy-tailed, Light-tailed, Bimodal, Normal with outlier(s); then cite the evidence you see.

Chart 1: lab experiment duration (minutes), n = 60



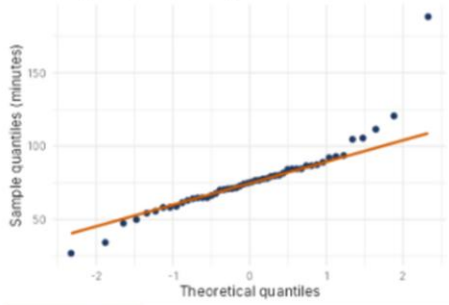
Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 2: commute time (minutes), n = 30



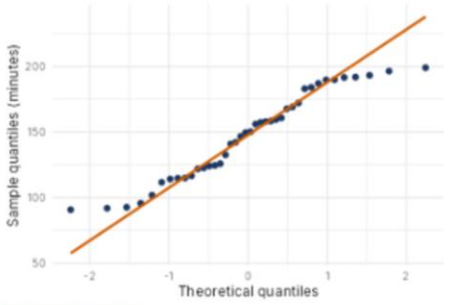
Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 3: daily study time (minutes), n = 50



Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 4: daily study time (minutes), n = 40



Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

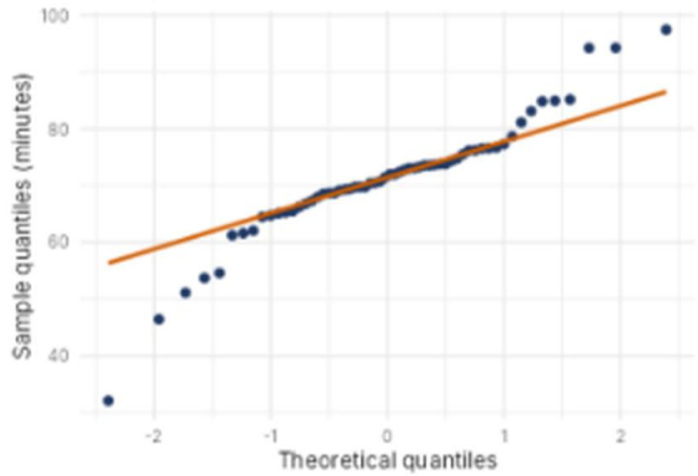
Part B (4 marks)

B1. Chart 3 departs from normality. Name the departure and state the remedy you would try before using mean-and-SD methods (for example log transform, square, split the data, robust methods). [2]

B2. Fill the blanks to draw a Q-Q plot with its reference line for a data frame df with a column value: ggplot(df, aes(sample = value)) + _____ + _____ [2]

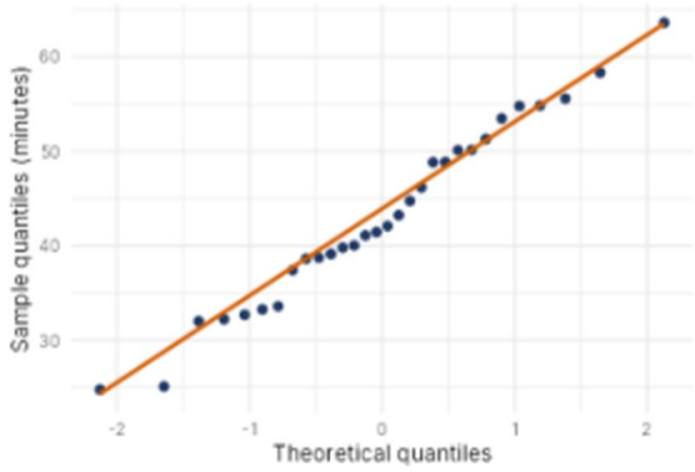
Part A

Chart 1: lab experiment duration (minutes), n = 60



Normal? (Yes / No)	No
Shape (from the list)	Heavy-tailed
Evidence in the plot	The middle of the plot (about -1 to +1) sits right on the red line, but both tails peel away from it in opposite directions. The low end falls below the line (points near 32, 46, 51, 54 where the line predicts roughly 56-63), and the high end rises above it (points near 94, 94, 97 versus a line value of about 85). This S-shaped pattern means the data have more extreme values on both sides than a normal distribution would give, so the tails are heavy.

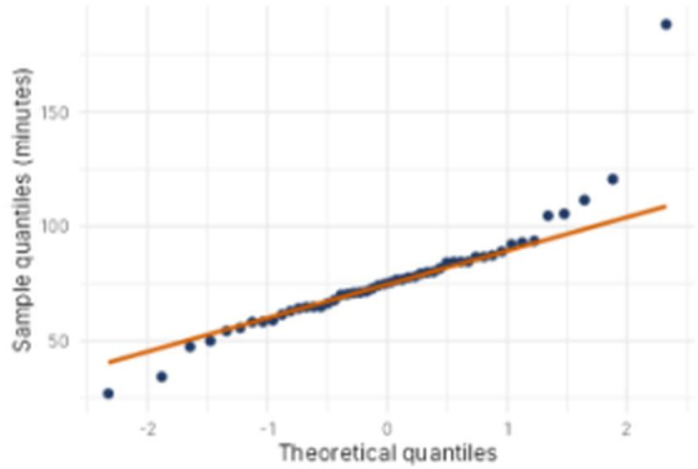
Chart 2: commute time (minutes), n = 30



Normal? (Yes / No)	Yes (approximately)
Shape (from the list)	Normal
Evidence in the plot	The points follow the red reference line closely from one end to the other, with no systematic curve, no S-shape and no isolated extreme point. The only small wobbles are the two lowest points, which sit slightly below the line. This is within

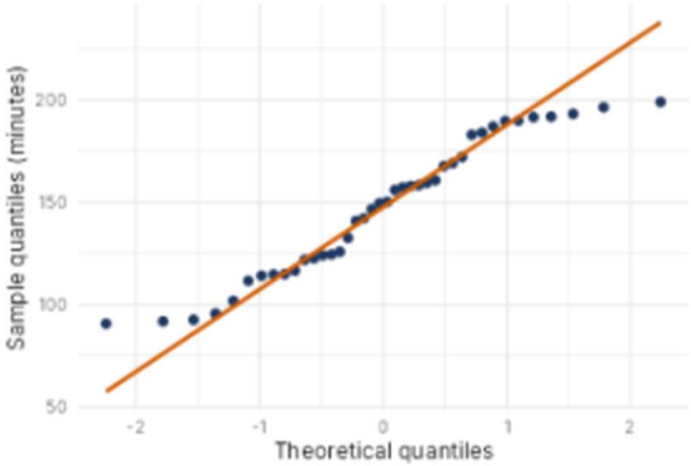
	normal sampling variation for n = 30, so the data are consistent with a normal distribution.

Chart 3: daily study time (minutes), n = 50



Normal? (Yes / No)	No
Shape (from the list)	Normal with outlier(s)
Evidence in the plot	Most of the points (roughly -1.5 to +1.2 on the x-axis) lie tightly along the line, so the bulk of the data is normal. One point at the top right (about 190 minutes) is far above the line and well separated from the rest, with the next-highest points near 105-120. The two lowest points also dip a little below the line. The single extreme value is what breaks normality, so the shape is normal with an outlier.

Chart 4: daily study time (minutes), n = 40



Normal? (Yes / No)	No
Shape (from the list)	Light-tailed
Evidence in the plot	The points form a reversed S-shape around the line. At the low end the points sit above the line and flatten out near 90-95 minutes, although the line predicts values down to about 60. At the high end the points sit below the line and flatten near 190-200 minutes, although the line predicts values above 230. The tails are therefore shorter than a normal distribution's, with no outliers, which is the signature of a light-tailed (platykurtic) distribution.

Part B

B1. Chart 3 departs from normality. Name the departure and state the remedy you would try before using mean-and-SD methods (for example log transform, square, split the data, robust methods).

Departure: Chart 3 is normal in the middle but has a single extreme high outlier (about 190 minutes) far above the red line, with a few mildly high values (105-120) before it.

Remedy: First check whether the extreme value is a data-entry or recording error by going back to the source. If it is an error, correct or remove it and document the decision. If it is a genuine value, do not just delete it. Either analyse the data with and without the outlier and report both results, or use robust methods that resist outliers (median and IQR or MAD instead of mean and SD, a trimmed or winsorized mean, or a rank-based test). A log transform is a secondary option, because it pulls in high values but is not needed when only one point is extreme and the rest of the data are already normal.

B2. Fill the blanks to draw a Q-Q plot with its reference line for a data frame df with a column value: ggplot(df, aes(sample = value)) + _____ + _____

Answer:

```
ggplot(df, aes(sample = value)) + stat_qq() + stat_qq_line(colour = "red")
```

Blank 1: stat_qq() (draws the sample points against theoretical normal quantiles)

Blank 2: stat_qq_line() (adds the reference line; colour = "red" is optional and matches the red line in the charts)